EARTHWORM PLATFORM SPECIFICATION

AI-Powered Market Linkages & Price Discovery Network

PROBLEM STATEMENT: SIH26132

TEAM: GREEN CODERS

🌿 SIH26132 Innovation Overview: Solving the Operational Gaps of e-NAM & Agmarknet

EARTHWORM transforms agricultural marketing for India's 86% smallholder farmers (<2.5 acres). Unlike legacy government portals that only record historical modal prices, EARTHWORM provides **transaction-level execution**: calculating localized net farmgate prices (per-km freight deducted), eliminating distant buyer trust deficits with computer vision quality lotting, aggregating smallholder logistics via shared ride-pooling, preventing panic distress dumping through holding economics, and providing statutory dispute redressal.

+35%	64%	+₹27,300	100%
Net Take-Home Earnings	Freight Saved (₹2.2k ➔ ₹800)	Cold Storage Glut Gain	DPDP Act Zero-PII

CORE PLATFORM ARCHITECTURE & SCREEN-BY-SCREEN FEATURES

📱 1. Dedicated Farmer Authentication — SCREEN 00

Zero-friction mobile login tailored for rural farmers with automated SIM OTP verification and fast-track SIH demo mode.

- ✓ **OTP Verification:** 4-digit automated authentication (+91 98421 78902).
- ✓ **DPDP Act Compliance:** Farmer phone/identity hashed to private alias `#EW-APP-8842`.
- ✓ **1-Touch Evaluation Mode:** Pre-filled model Tamil Nadu farm profile for judges.
- ✓ **Vernacular Voice AI:** 1-touch toggle between IN தமிழ் and GB English.

🗺️ 2. In-Registration GIS Land Drawing — SCREEN 01

Interactive satellite map embedded directly into registration, allowing farmers to draw their field boundaries.

- ✓ **Touch Polygon Tool:** Draw custom farm boundary corners on Leaflet satellite map.
- ✓ **Auto-Acreage Engine:** Turf.js calculates exact digital acreage (e.g. 2.50 ac).
- ✓ **ISRO Bhuvan Ingestion:** Displays real-time soil classification (Red Loam, pH 6.9).
- ✓ **Sentinel-2 NDVI Vigor:** 10m resolution vegetative health baseline (0.74 score).

🏠 3. Problem-Solution Master Dashboard — SCREEN 02

Centralized operational hub directly addressing SIH26132 requirements with transparent net earnings and glut alerts.

- ✓ **Net Price Realization Meter:** Mandi price (₹27.50) minus per-km freight (-₹1.30) = Net ₹26.20/kg.
- ✓ **Break-Even Cost Safety Line:** Displays unit cost (₹10.20/kg) vs profit (+₹16.00/kg).
- ✓ **Mandi Glut Surge Banner:** Real-time +42% arrival surge alert with holding advice.
- ✓ **Direct Institutional Contracts:** 1-click delivery matching with verified food processors.

📊 4. "What-If" AI Financial Simulator — SCREEN 03

Interactive economic modeling engine allowing farmers to simulate market shocks and download official loan dossiers.

- ✓ **Unit Production Calculator:** ₹38,500 working capital on 2.5 acres yields 37.5 Tons.
- ✓ **Quick Scenario Chips:** *"What if Tomato?"*, *"What if -30% Crash?"*, *"What if Onion?"*.
- ✓ **Gemini 1.5 Flash Math:** Calculates gross vs net returns under crashed market scenarios.
- ✓ **KCC Loan Dossier:** 1-click printable PDF financial dossier (#KCC-TN-4820) for 4% crop loans.

EARTHWORM PLATFORM SPECIFICATION (CONTD.)

Post-Harvest Logistics, Assaying & Redressal Modules

SIH26132

MODULES 04 TO 08

📸 5. AI Produce Quality Assaying — SCREEN 04

Computer vision produce scanner that breaks e-NAM's remote buyer trust deficit right at the farmgate.

- ✓ **3-Second Camera Scan:** Segments harvest crates via smartphone camera reticle.
- ✓ **Chromatic Uniformity:** Assays red turning ripeness index (94.8% score).
- ✓ **Morphometric Sizing:** Computes average fruit diameter (66 ± 3 mm Grade A).
- ✓ **Digital Lot Passport:** Mints verifiable, tamper-evident **#LOT-TN-902** with QR code.

❄️ 6. Distress Sale vs. Storage Math — SCREEN 05

Interactive economic holding optimizer preventing roadside distress dumping during peak seasonal gluts.

- ✓ **Dynamic Holding Slider:** Adjust harvest volume (2,000 kg) & holding weeks (1 to 8 weeks).
- ✓ **Distress Glut Payout:** Compares panic dump (₹20,000 at ₹10/kg) vs post-glut rate (₹25/kg).
- ✓ **Net Extra Surplus:** Recommends optimal sale week unlocking **+ ₹27,300 net gain**.
- ✓ **AIF Warehouse Booking:** Direct reservation with 3% Agriculture Infrastructure Fund subsidy.

🚚 7. Shared Ride-Pooling (பங்கு வண்டி) — SCREEN 06

Less-Than-Truckload (LTL) shared logistics aggregator slashing freight expenses for smallholders.

- ✓ **64% Freight Reduction:** Drops mini-truck hire from ₹2,200 solo to ₹800 per farmer.
- ✓ **Village Hub Route Matching:** Aggregates loads from Polur/Chengam to Koyambedu & APMCs.
- ✓ **Capacity Tracking:** Real-time available truck space (700 kg - 1,200 kg slots).
- ✓ **DPDP Virtual Telephony:** Driver & farmer communicate via masked proxy (`044-6902-XXXX`).

⚠️ 8. Mandi Supply Glut Danger Radar — SCREEN 07

Early morning predictive radar forecasting wholesale price drops and providing direct institutional market linkages.

- ✓ **6:00 AM Glut Forecast:** Identifies +42% arrival surges and warns of -18% price drop by 11:00 AM.
- ✓ **Localized Net Realization Index:** Ranks 3 nearby mandis with per-km transit costs deducted.
- ✓ **Direct Institutional Buyers:** Verified contracts with Annapoorna Food Processing (₹28/kg).
- ✓ **Zero Commission:** Eliminates 15%-25% intermediary commissions via direct buyer handshakes.

⚖️ 9. Statutory Dispute Redressal & Google Gemini Vernacular AI Assistant — SCREEN 08

Empowers smallholders with statutory 24-hour APMC legal dispute arbitration and 24/7 conversational agricultural intelligence.

- ✓ **Digital Dispute Docket:** Issues trackable docket (`#EW-GR-4820`) for delayed buyer payments.
- ✓ **Statutory 24-Hr APMC SLA:** Enforces mandatory buyer settlement within 24 hours of weighment.
- ✓ **Government Gateways:** Direct links to e-NAM Grievance Cell, CPGRAMS, & Toll-Free 1800-180-1551.
- ✓ **Google Gemini 1.5 Flash:** Multi-crop price intelligence (Onion, Tomato, Paddy, Groundnut).
- ✓ **Zero-Key Cloud Proxy:** 100% free for farmers under Digital India DPI / AgriStack.
- ✓ **Offline Smart AI Engine:** Instant local mathematical fallback during rural network outages.

E-NAM VS. EARTHWORM SOLUTION COMPARISON

Evaluation Dimension	Legacy Government Portals (e-NAM / Agmarknet)	EARTHWORM Innovation (SIH26132)
Price Reporting	Opaque gross mandi rates (No freight or commission deducted)	**Transparent Net Realization Index** (Per-km freight deducted)
Smallholder Logistics	Zero transport support (Farmer pays ₹2,200 solo truck hire)	**Shared Ride-Pooling (பங்கு வண்டி)** (64% savings: ₹800/farmer)
Quality Verification	Physical APMC laboratory testing (Takes 24-48 hours)	**On-Farm AI Computer Vision Assaying** (#LOT-TN-902 in 3 seconds)

Evaluation Dimension	Legacy Government Portals (e-NAM / Agmarknet)	EARTHWORM Innovation (SIH26132)
Seasonal Supply Gluts	No decision math (Farmers dump produce at ₹5/kg)	**Liquidity-Adjusted Cold Storage Window** (+₹27,300 net gain)
Dispute Settlement	Informal middleman control; delayed settlement (> 7 days)	**Statutory 24-Hr APMC Arbitration** & CPGRAMS digital dockets
Farmer Privacy	Phone numbers shared publicly with commission agents	**DPDP Act 2023 Masked Virtual Calling** (044-6902-XXXX)